

Solution Code - Module 3-Core Java - LeapYearChecker

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q04_LeapYearChecker.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q04_LeapYearChecker {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter a year: ");
7 |         if (scanner.hasNextInt()) {
8 |             int year = scanner.nextInt();
9 |             boolean isLeap = false;
10 |
11 |             if (year % 4 == 0) {
12 |                 if (year % 100 == 0) {
13 |                     if (year % 400 == 0) {
14 |                         isLeap = true;
15 |                     } else {
16 |                         isLeap = false;
17 |                     }
18 |                 } else {
19 |                     isLeap = true;
20 |                 }
21 |             } else {
22 |                 isLeap = false;
23 |             }
24 |
25 |             if (isLeap) {
26 |                 System.out.println(year + " is a leap year.");
27 |             } else {
28 |                 System.out.println(year + " is not a leap year.");
29 |             }
30 |         } else {
31 |             System.out.println("Error: Invalid year entered.");
32 |         }
33 |         scanner.close();
34 |     }
35 | }
```